AIORI-3 CYBERSECURITY HACKATHON | PS-011

ArgusTrace

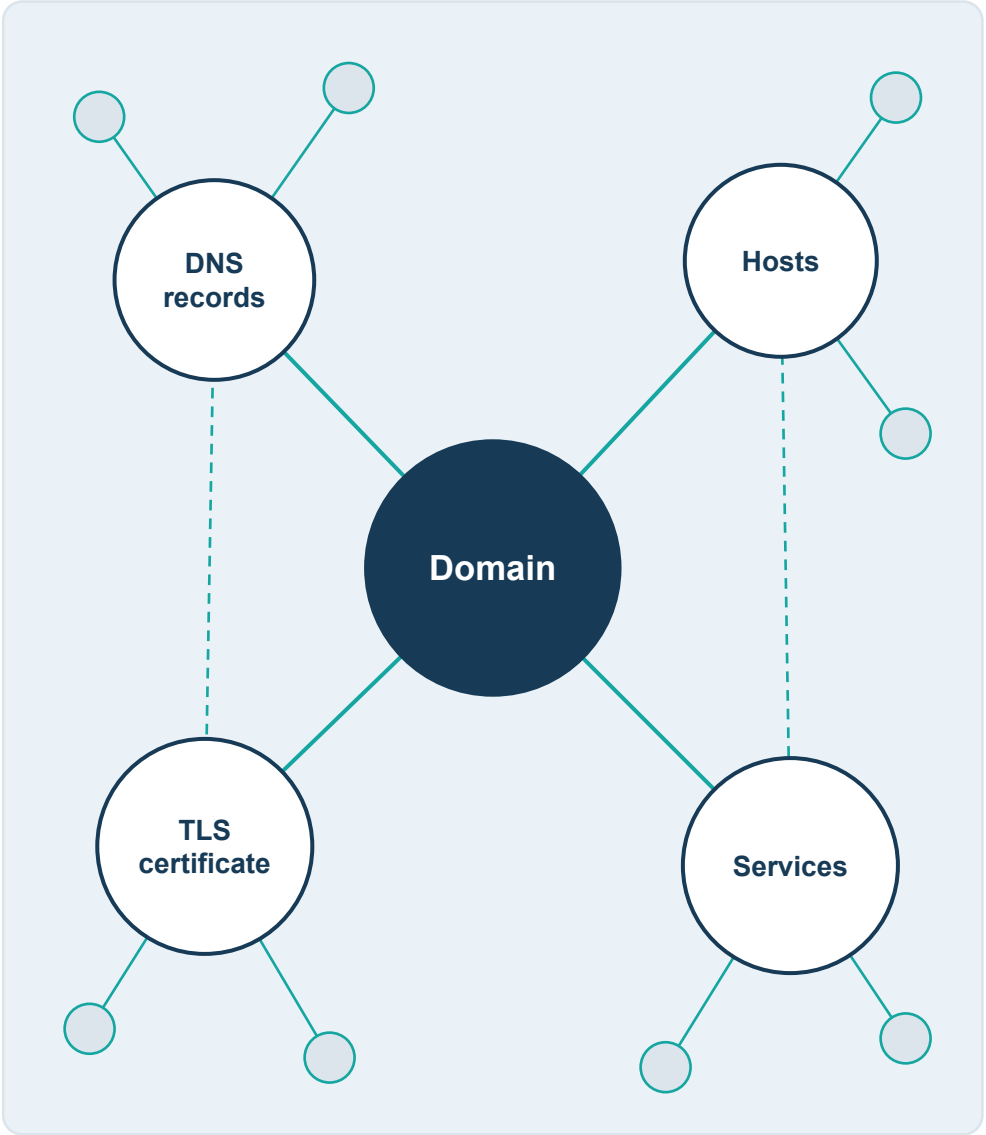
Authorised Attack-Surface Inventory and Change Monitor

Discover. Monitor. Detect. Protect.

TEAM VANTANEX

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Abstract and Problem Description

A. What is PS-011?

PS-011, Authorised Attack-Surface Inventory and Change Monitor, is a cybersecurity problem statement on maintaining an inventory of approved digital assets and detecting meaningful changes in their DNS records, TLS certificates and exposed services over time. The objective is repeatable, traceable, reviewable monitoring through scheduled observations and baseline comparison.

B. Abstract

Cloud infrastructure, websites, domains and network services change frequently. Without regular monitoring, exposed assets may go unnoticed, TLS certificates may expire, unexpected services may become reachable, and DNS records may point to decommissioned cloud resources.

ArgusTrace proposes an authorised attack-surface inventory and change-monitoring system for domains, DNS records, TLS certificates and exposed services. It will set a baseline inventory, run scheduled checks, identify unexpected changes and generate alerts and reports.

The proposed system will be evaluated with organiser-provided seeded issues and false-positive analysis in a controlled, approved lab environment.

C. Problem Description

1. Incomplete Asset Inventory

Organisations may lack an updated record of their exposed digital assets.

2. Expired TLS Certificates

Expired certificates can affect secure communications and service reliability.

3. Unexpected Open Ports

Newly exposed services may go unnoticed without regular monitoring.

4. Potential Dangling DNS Records

DNS records may point to decommissioned cloud resources.

5. Undetected Infrastructure Changes

Manual checks make it difficult to track changes consistently.

6. Lack of Actionable Alerts

Security teams need clear findings and evidence to investigate potential issues.

D. Problem Objective: Develop a lightweight monitoring system that identifies exposed assets and detects changes across an authorised lab scope.

Background and Technical Context

Attack Surface Management (ASM) is the process of identifying, cataloguing and monitoring an organisation's externally exposed digital assets.

1 DNS Monitoring

Collect supported DNS records, including A, AAAA and CNAME, to identify changes and potential dangling DNS candidates.

2 TLS Certificate Monitoring

Inspect supported TLS certificate metadata and expiry dates to identify expired or soon-to-expire certificates.

3 Service Discovery

Check selected TCP ports to identify reachable services within the approved scope..

4 Baseline Inventory

Store an initial snapshot of known domains, DNS records, certificates and service observations.

5 Scheduled Change Detection

Repeat inventory checks at configured intervals and compare them with previous observations.

6 Asynchronous Network Operations

Use Python asyncio for concurrent I/O, which may improve monitoring efficiency. No measured gain is claimed before testing.

Project Objectives

- Monitor up to 20 authorised lab domains and 10 hosts.
- Maintain a structured inventory of assets and observations.
- Detect unexpected services and relevant DNS/TLS changes.
- Generate alerts and inventory change reports.
- Evaluate seeded issues, false positives and missed detections.

Safety note

All monitoring will remain restricted to written, organiser-approved targets.

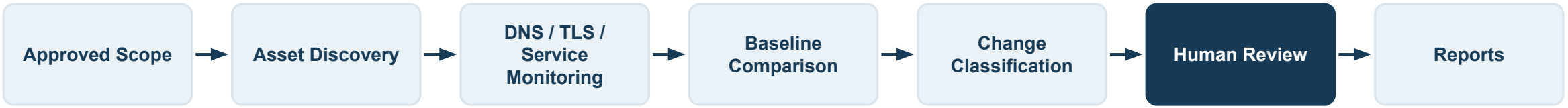
Literature and RFC Review

Reference	Main concept documented by the source	Relevance to ArgusTrace (our proposal)
RFC 9293 — Transmission Control Protocol (TCP) rfc-editor.org/rfc/rfc9293.html	Defines TCP operation and connection behaviour.	Background for the proposed TCP reachability checks.
RFC 4632 — Classless Inter-Domain Routing (CIDR) rfc-editor.org/rfc/rfc4632.html	Explains IP addressing and prefix aggregation.	Relevant to IP-range representation, if approved CIDR processing is implemented.
RFC 5280 — Internet X.509 PKI Certificate and CRL Profile rfc-editor.org/rfc/rfc5280.html	Provides the X.509 certificate profile, including validity periods (not the whole TLS protocol).	Relevant to certificate metadata and expiry checks.
RFC 1034 — Domain Names: Concepts and Facilities datatracker.ietf.org/doc/html/rfc1034	Explains DNS concepts, architecture, domain names and resource records.	Background for DNS record collection and CNAME handling.
RFC 1035 — Domain Names: Implementation and Specification datatracker.ietf.org/doc/html/rfc1035	Describes DNS implementation and resource records used in name resolution.	Reference for record formats the DNS checker would work with.
Python asyncio Documentation docs.python.org/3/library/asyncio.html	Guidance on asynchronous programming and concurrent I/O.	Basis for proposed concurrent checks; any speed-up is to be measured.
OWASP Web Security Testing Guide (WSTG) owasp.org/www-project-web-security-testing-guide/	Structured security-testing guidance; its attack-surface identification page covers domains, DNS, services and certificates.	Informs which asset types to inventory. No compliance is claimed.
NIST Cybersecurity Framework 2.0 nist.gov/cyberframework	A framework for understanding and managing cybersecurity risk, including asset management.	Context for why asset inventories matter. No compliance is claimed.

Security principle: An unusual DNS record or reachable port is an observation requiring review. A potential dangling DNS record does not automatically prove that a resource is exploitable.

ArgusTrace — Proposed Monitoring System

ArgusTrace proposes to combine asset discovery, monitoring, inventory management, change detection and reporting in one workflow. **Expected outcome: a repeatable and auditable process for authorised attack-surface monitoring.**



1. Authorised Asset Discovery

- Accept approved domains, hosts and supported IP ranges
- Validate targets against the approved scope
- Resolve supported DNS records
- Organise findings into a structured inventory

2. DNS, TLS and Service Monitoring

- Collect DNS records and track relevant changes
- Inspect TLS certificate metadata and expiry
- Perform approved service checks
- Store results with timestamps and evidence

3. Baseline and Change Detection

- Save an initial inventory snapshot
- Run recurring checks on a configured schedule
- Compare new observations with previous data
- Identify new, removed or changed services and records

4. Alerting and Reporting

- Flag unexpected services
- Report expired TLS certificates
- Identify potential dangling DNS candidates for manual validation
- Generate inventory and change reports; plan JSON/CSV export

5. Testing and Evaluation

- Test against organiser-provided seeded issues
- Record detections, false positives and missed detections
- Produce a final technical report and demonstration

Candidate checks — pending organiser approval

- TCP reachability of an approved host and port
- TLS certificate validity, where TLS applies
- HTTP response/status of an approved web endpoint

Basic banner identification on an approved port, only if explicitly permitted

Solution Optimization

Area	Basic / Manual Approach	Proposed ArgusTrace Approach	Expected Benefit	Limitation or Verification Requirement
Asset Inventory	Separate, manually updated lists	Centralised structured inventory	One reference list of approved assets	Completeness depends on the approved scope and quality of observations
Port Inspection	Individual sequential checks	Concurrent asynchronous checks using Python asyncio, if implemented	Potentially more efficient collection of approved checks	Actual performance improvement must be measured
DNS Monitoring	Occasional manual lookups	Repeatable DNS collection and change tracking	Relevant record changes become easier to identify	DNS responses may vary because of caching, propagation or temporary failures
TLS Monitoring	Manual certificate inspection	Automated certificate metadata and expiry checks	Validity dates can be tracked consistently	Certificate validity checking is not a full TLS security assessment
Change Detection	Manual snapshot comparison	Automated baseline comparison	Changes can be identified systematically	Initial baseline must be reviewed and may not be a complete inventory
Scan Scheduling	Manually initiated checks	Configurable scheduled monitoring	Repeatable observations at defined intervals	Not real-time monitoring; depends on the configured interval
Alerting	Findings reviewed separately	Centralised change alerts with evidence	Findings are easier to investigate	A reviewer must still decide whether an alert is meaningful
Evaluation	Informal verification	Seeded-issue testing and false-positive / missed-detection analysis	Repeatable evaluation of the proposed system	Quality depends on the test cases and the approved lab environment

Expected Benefits (proposed)

- Reduce repeated manual monitoring effort
- Improve consistency across inventory checks
- Identify relevant changes between scheduled scans
- Make findings easier to investigate with evidence
- Support repeatable testing and evaluation

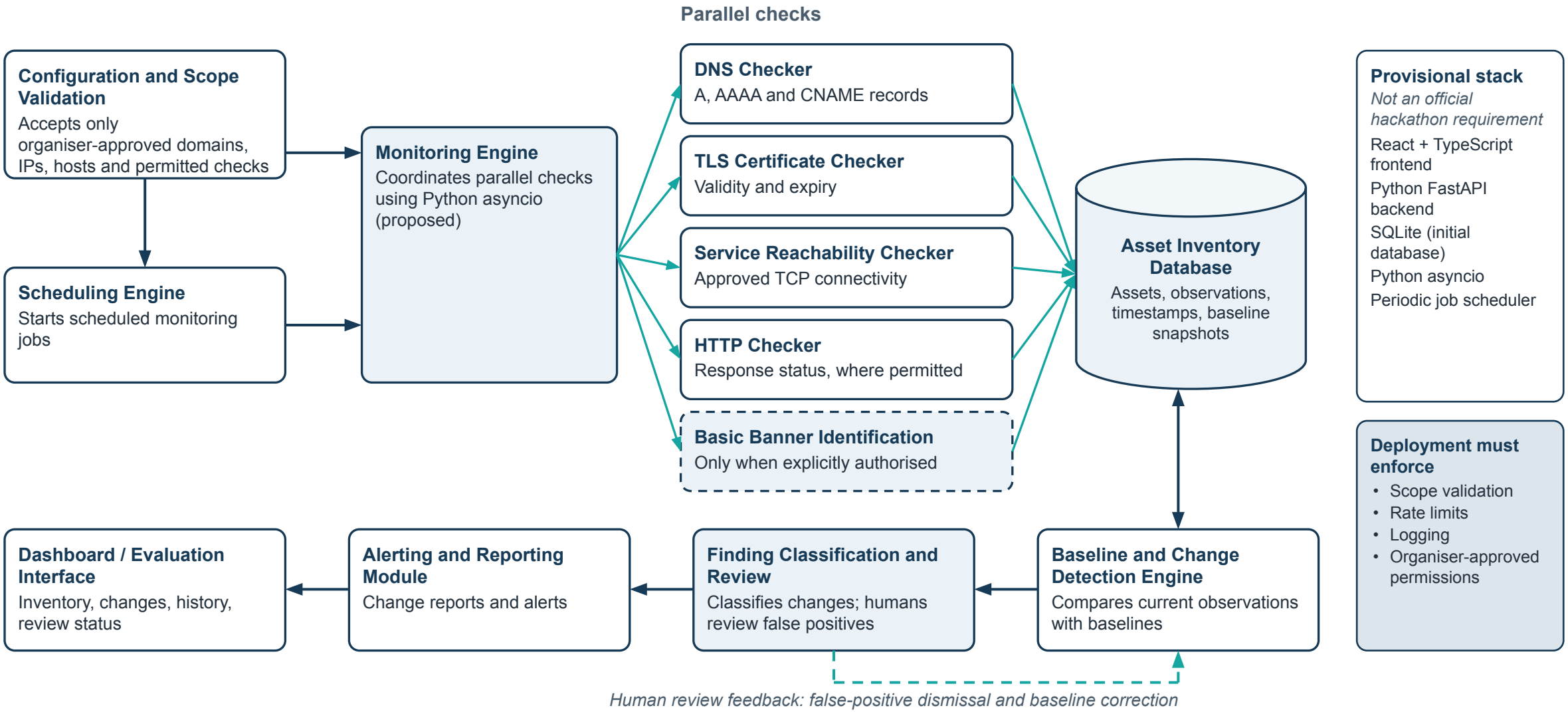
Proposed improvements are not measured results. No time savings, detection accuracy or performance figures are claimed before testing.

Scope control: only organiser-approved targets are checked.

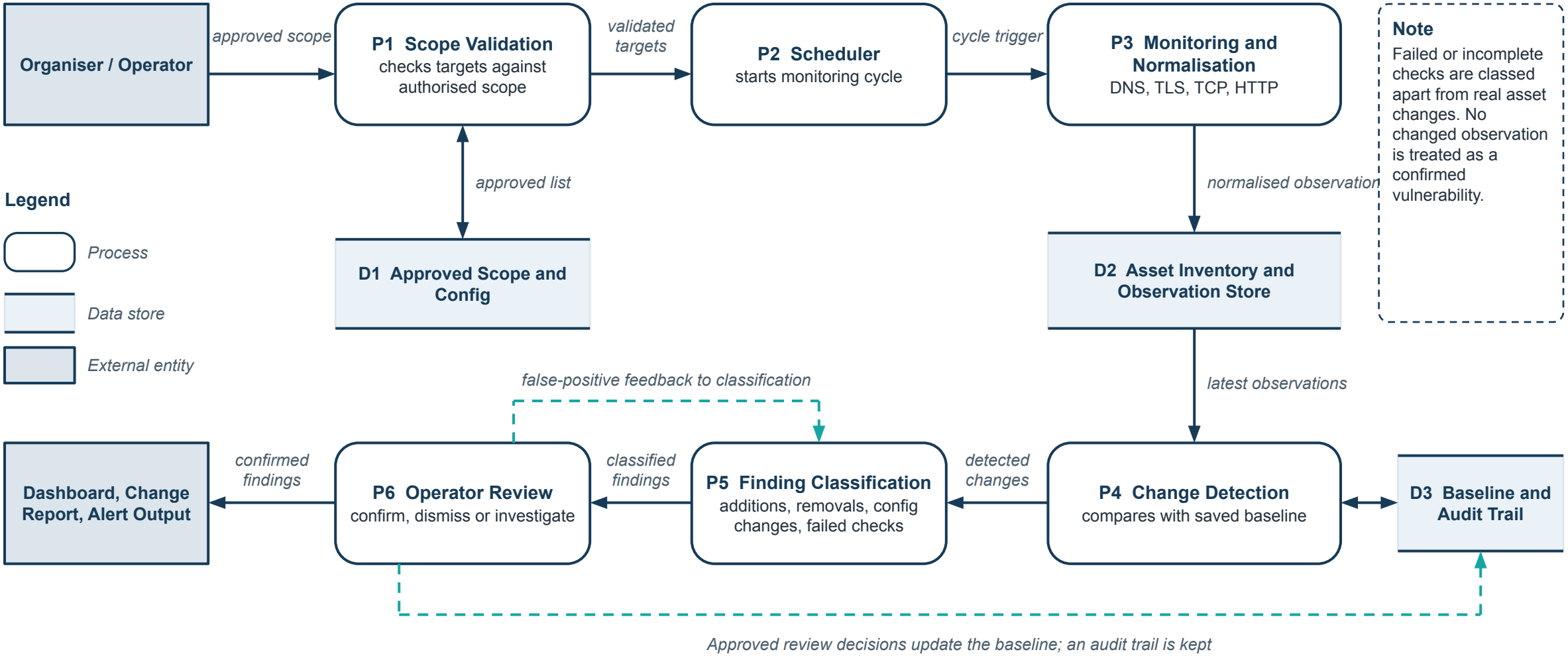
False-positive handling: findings may require human verification; automated observations are not automatically confirmed vulnerabilities.

Solution Architecture and Design

Proposed Architecture



Data Flow Diagram (DFD)



Deployment Timeline

Proposed ArgusTrace technical work plan

Phase 1 — Foundation and Asset Discovery
Days 1–3

- Set up the development environment and repository.
- Create configuration (`src/config.py`) and approved-target validation.
- Implement DNS resolution and supported IP/CIDR processing (`src/network.py`).
- Develop the initial TCP port-checking engine (`src/async_scanner.py`).
- Verify functionality using controlled lab targets.

Phase 2 — Monitoring and Change Detection
Days 4–5

- Implement TLS certificate metadata collection and expiry checks.
- Develop structured inventory and baseline storage (`src/exporter.py`).
- Implement inventory comparison and change classification.
- Configure scheduled monitoring.
- Test detection of changes between inventory snapshots.

Phase 3 — Alerts, Reporting, and Validation
Days 6–7

- Implement alerts for relevant changes and potential issues.
- Develop inventory and change reports with JSON/CSV export handlers.
- Build the dashboard interface (`src/app.py` and `templates/index.html`).
- Test seeded expired-certificate, unexpected-service, and potential dangling-DNS cases.
- Document false positives and missed detections (validated via 25 passing pytest tests).
- Prepare the final technical report and demonstration setup.

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[1] IETF. RFC 1034 — Domain Names: Concepts and Facilities.
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[11] AIORI. AIORI-3 (programme and timeline page).
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[12] AIORI. PS-011 — Authorised Attack-Surface Inventory and Change Monitor (problem statement), linked from portal.aiori.in/aiori-3/hackathon/.
https://drive.google.com/file/d/12OzcdPKMR0XC6m83xd9tF-T5fWxWPf4a/view

Thank You !

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